

# aflevering 7

2025-04-01

```
firmaer<-read.csv("/Users/asgeraircool/rdata/Firmaer.csv")
head(firmaer)
```

```
##   Obs  omsaet  vaerdi sektor
## 1    1 9.08171 8.39344 detail
## 2    2 7.37838 6.75926 detail
## 3    3 7.71735 6.43455 detail
## 4    4 8.79710 7.38275 detail
## 5    5 9.74824 8.33759 detail
## 6    6 8.53307 8.14381 detail
```

```
unique(firmaer$sektor)
```

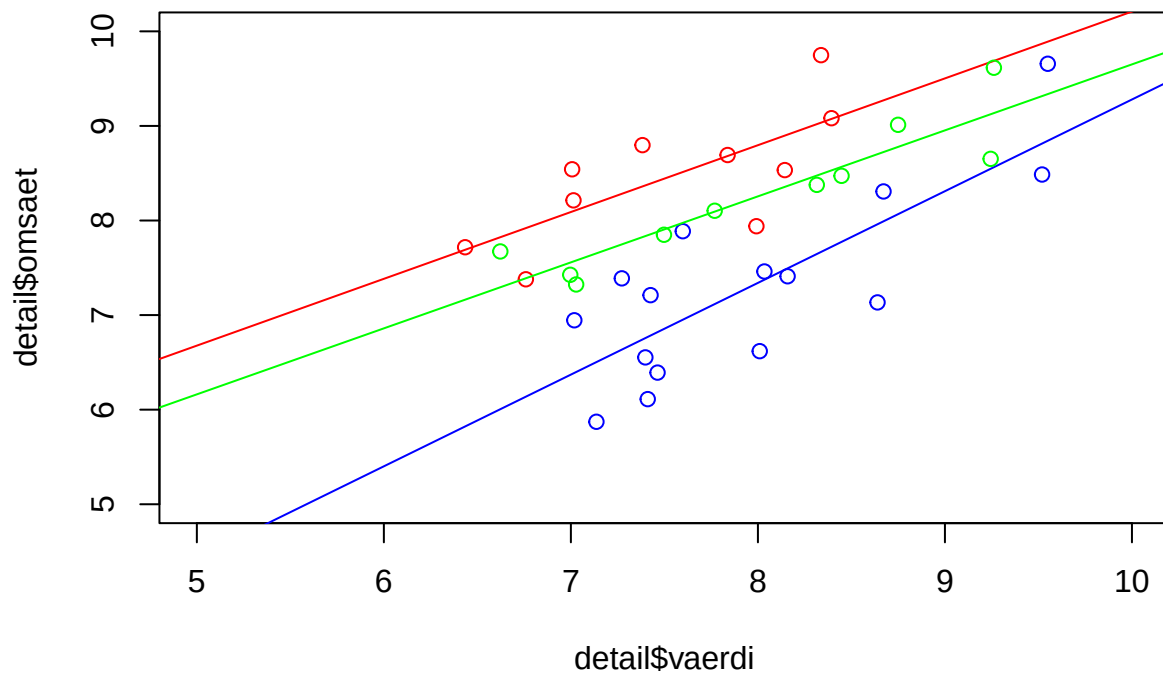
```
## [1] "detail"  "energi"  "industri"
```

```
detail<-subset(firmaer,subset=sektor=="detail")
energi<-subset(firmaer,subset=sektor=="energi")
industri<-subset(firmaer,subset=sektor=="industri")
```

```
##1
```

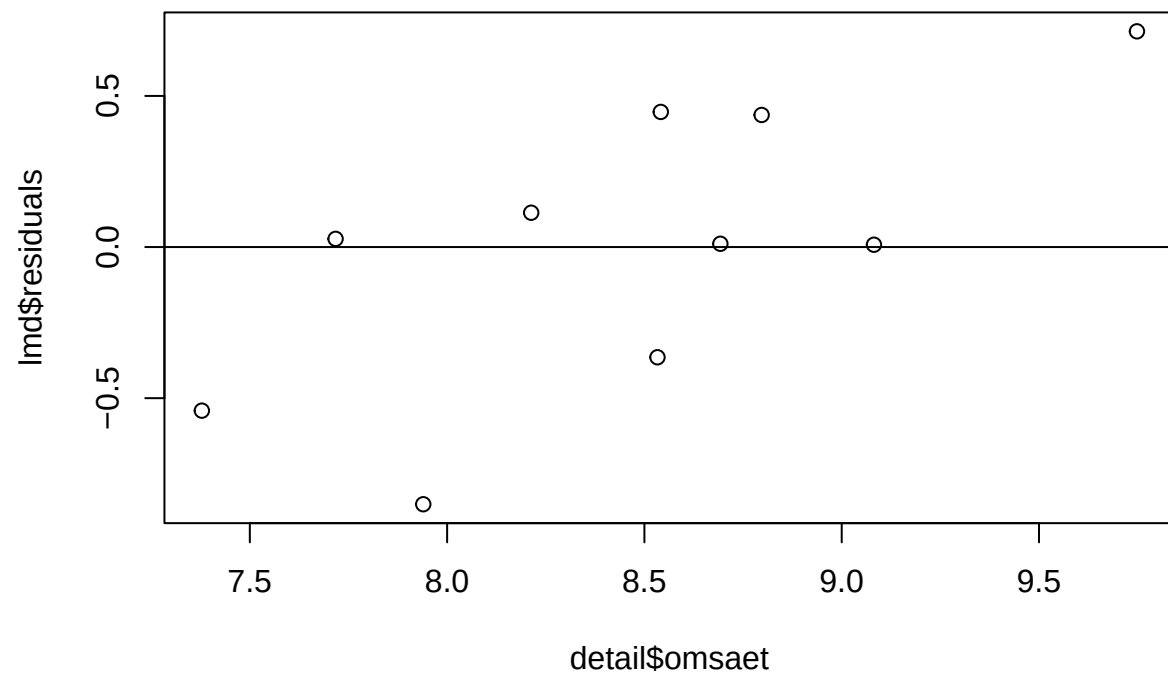
```
lmd<-lm(omsaet~vaerdi,data=detail)
lme<-lm(omsaet~vaerdi,data=energi)
lmi<-lm(omsaet~vaerdi,data=industri)

plot(detail$vaerdi,detail$omsaet,col="red",xlim=c(5,10),ylim=c(5,10))
abline(lmd,col="red")
points(energi$vaerdi,energi$omsaet,col="blue")
abline(lme,col="blue")
points(industri$vaerdi,industri$omsaet,col="green")
abline(lmi,col="green")
```

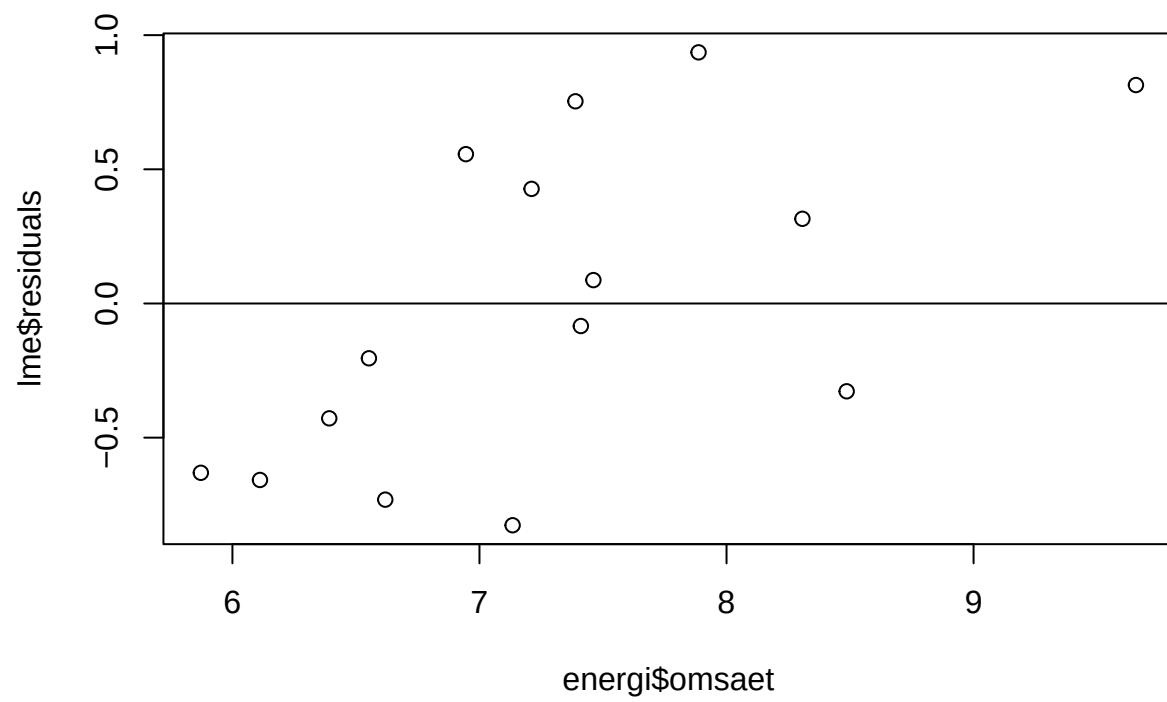


det ser rimeligt ud at antage en lineær sammenhæng for at tjekke efter tegne vi residualplot

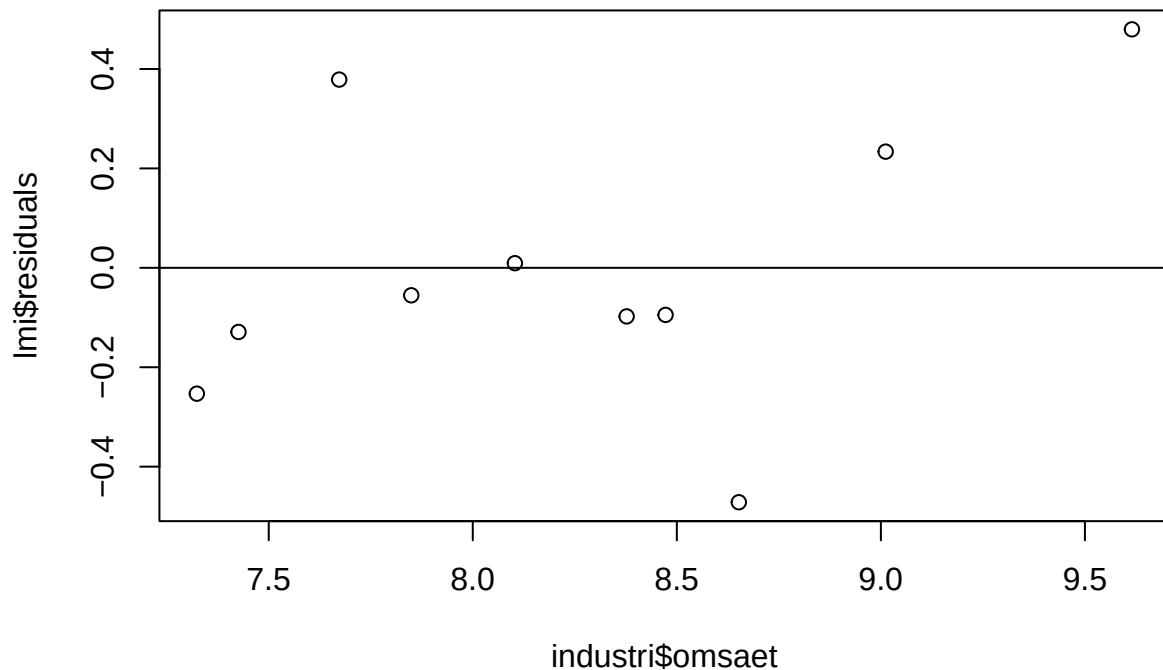
```
plot(lmd$residuals~detail$omsaet)  
abline(0,0)
```



```
plot(lme$residuals~energi$omsaet)  
abline(0,0)
```



```
plot(lmi$residuals~industri$omsaet)  
abline(0,0)
```



igen ser residualerne rimelige ud uden systematisk afvigelse

altså starter vi i modellen  $M_0 : O_{i,k} \sim N(\alpha_i, \beta_i * V_{i,k}, \sigma_i^2)$  for  $i \in \{detail, energi, industri\}$  for  $k \in \{1, \dots, n_i\}$

##2  $H_0 : \sigma_i^2 = \sigma_0^2$  for  $i \in \{detail, energi, industri\}$  vi kører en bartletts test for at tæste hypotesen

```
bartlett.test(list(lmd, lme, lmi))
```

```
##
## Bartlett test of homogeneity of variances
##
## data: list(lmd, lme, lmi)
## Bartlett's K-squared = 4.0517, df = 2, p-value = 0.1319
```

p værdien er ikke kritisk så vi går videre med ens varians ##3 vi opskriver de rigtige nære modeller og kører anova for at se om det er rimligt at reducere modellen  $M_1 := O_{i,k} \sim N(\alpha_i, \beta_i * V_{i,k}, \sigma_0^2)$  for  $i \in \{detail, energi, industri\}$  for  $k \in \{1, \dots, n_i\}$

$$M_2 : O_{i,k} \sim N(\alpha_i, \beta * V_{i,k}, \sigma_0^2) \text{ for } i \in \{detail, energi, industri\} \text{ for } k \in \{1, \dots, n_i\}$$

vi tester hypotesen om at fi kan reducere fra m1 til m2 med anova

```
lmm1<-lm(omsaet~sektor*vaerdi,data=firmaer)
lmm2<-lm(omsaet~sektor+vaerdi,data=firmaer)
anova(lmm2,lmm1)
```

```
## Analysis of Variance Table
##
## Model 1: omsaet ~ sektor + vaerdi
## Model 2: omsaet ~ sektor * vaerdi
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      31 8.2962
## 2      29 7.9097  2   0.38657 0.7087 0.5006
```

p værdien er ikke kritisk da kan det godt antages at hældningen er den samme ##4 vi kører summary på den lineære model

```
summary(lmm2)
```

```
##
## Call:
## lm(formula = omsaet ~ sektor + vaerdi, data = firmaer)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.90102 -0.36075 -0.01226  0.37861  1.06092
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.3275     0.8486   2.743  0.0100 *
## sektorenergi   -1.5139     0.2163  -6.998 7.44e-08 ***
## sektorindustri -0.5912     0.2370  -2.495  0.0181 *
## vaerdi          0.8150     0.1106   7.370 2.68e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5173 on 31 degrees of freedom
## Multiple R-squared:  0.7465, Adjusted R-squared:  0.7219
## F-statistic: 30.42 on 3 and 31 DF,  p-value: 2.299e-09
```

vi aflæser at p værdien for at  $\beta = 0$  som er meget kritisk altså kan vi ikke antage at hældningen er lig 0  
##5 vi udregner konfidensintervallet

```
df<-lmm2$df.residual
s<-sigma(lmm2)
upper<-sqrt((df*s^2)/qchisq(0.025,df))
lover<-sqrt((df*s^2)/qchisq(1-0.025,df))
lover
```

```
## [1] 0.4147375
```

```
upper
```

```
## [1] 0.6877669
```

altså er 95% konfidensintervallet for  $\sigma_0$  [0.4147375 ,0.6877669]

##6

vi aflæser fra confint

```
confint(lmm2)
```

```
##                2.5 %      97.5 %  
## (Intercept)    0.5967555  4.0582560  
## sektorenergi  -1.9551102 -1.0726982  
## sektorindustri -1.0745116 -0.1079832  
## vaerdi         0.5894214  1.0404821
```

da har vi at 95% konfidensintervallet for  $\beta$  er [0.5894214, 1.0404821]